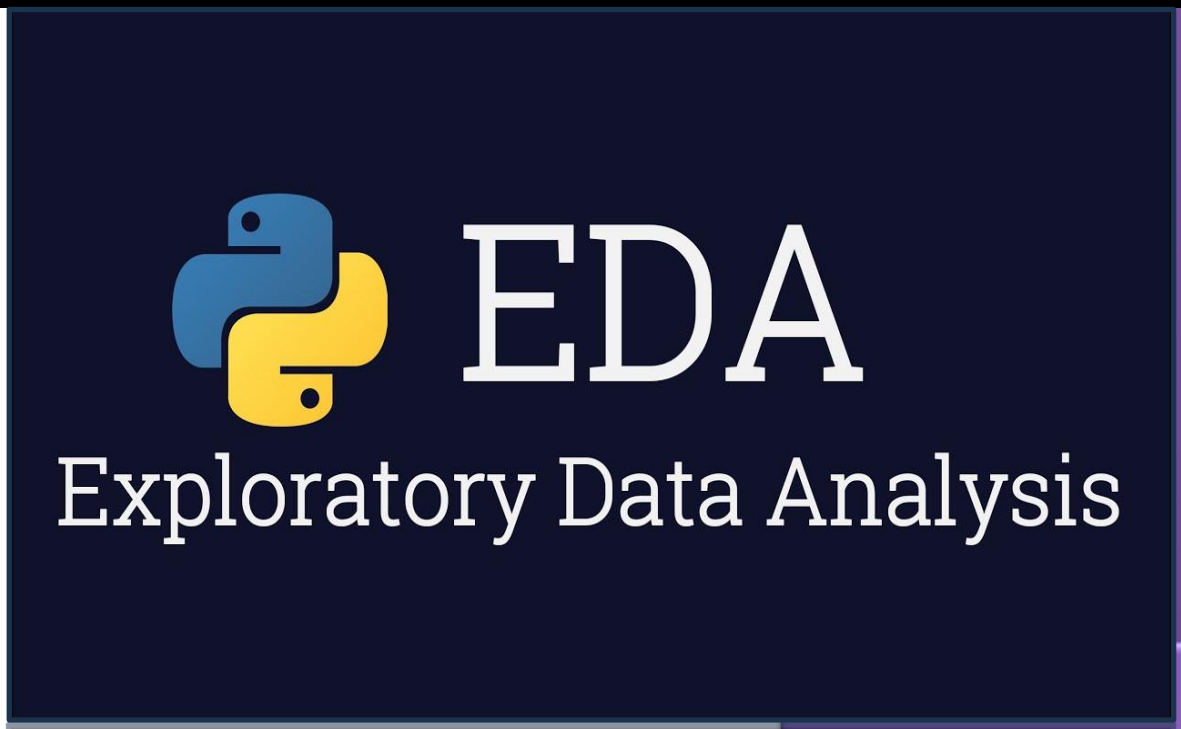
2024

EDA Report for AeroFit Treadmill Buyer Profile



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EDA Report for AeroFit Treadmill Buyer Profile

Introduction:

The report provides insights into the characteristics of customers purchasing various models of AeroFit treadmills (KP281, KP481, and KP781) using EDA (exploratory data analysis) in python language. The goal is to understand the distribution of key customer characteristics and how they differ across the types of treadmills. This analysis will help AeroFit provide better recommendations to new customers based on their demographic and behavioral attributes.

Dataset Overview:

The dataset provided contains information about individuals who purchased a treadmill from AeroFit stores during the prior three months.

Data Composition:

The dataset consists of 9 features and 180 customer records.

It includes a mix of numerical and categorical data.

Column Types:

3 categorical columns: Product, Gender, Marital Status.

6 numerical columns: Age, Education, Usage, Fitness, Income, and Miles.

It includes the following features:

- Product: Treadmill model purchased (KP281, KP481, or KP781)
- Age: Age of the customer in years
- Gender: Gender of the customer (Male/Female)
- Education: Education level in years
- Marital Status: Whether the customer is single or partnered.
- Usage: Average number of times the customer plans to use the treadmill each week
- Fitness: Self-rated fitness level on a scale from 1 (poor shape) to 5 (excellent shape)
- Income: Annual income in US dollars
- Miles: Average number of miles the customer expects to walk/run each week

Data Exploration and Processing:

Importing Data:

The dataset was loaded into a pandas Data Frame using Python. The file used for analysis is named aerofit_treadmill_data.csv.

Shape and Size of the Data Frame:

The dataset consists of 180 rows (the number of customers) and 9 columns showing the features of dataset.

Data Types:

The dataset has 9 columns with their corresponding data types:

- Product: object (categorical data for treadmill models: KP281, KP481, KP781)
- Age: int64 (numerical, whole numbers representing customer age)
- Gender: object (categorical, Male/Female)
- Education: int64 (numerical, years of education)
- Marital Status: object (categorical, Single/Partnered)
- Usage: int64 (numerical, weekly treadmill usage frequency)
- Fitness: int64 (numerical, self-rated fitness level on a 1-5 scale)
- Income: int64 (numerical, annual income in USD)
- Miles: int64 (numerical, expected weekly miles to run)

Missing Values:

The data has no missing values.

Duplicates:

The dataset has no duplicates.

Dataset Readiness:

The data appears clean and well-structured for exploratory data analysis (EDA).

Statistical Summary:

The statistical summary of the numerical features shows:

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

Table 1: Statistical Summary

Customer Age:

The table shows that Age ranges from 18 to 50 years with an average of 28.8 years and majority of customers are under 33 years (75th percentile).

Education:

Education ranges from 12 to 21 years, with an average of 15.6 years and most customers have a minimum of 16 years of education (median).

Treadmill Usage:

Customers use treadmills on average 3.5 times per week, with a max of 7 times.

Fitness Level:

Fitness is self-rated between 1 to 5 and the average fitness level is 3.3, indicating moderate fitness overall.

Income:

The annual income ranges from $29,562 to $104,581, with an average of $53,719.6. Income has significant variation as seen in the standard deviation.

Miles Expected:

Miles range from 21 to 360, with an average of 103.2 miles. High variability suggests diverse fitness goals and treadmill usage.

Categorical Features

Product:

There are 3 product types, and KP281 is the most popular, with 44% (80/180) customers purchasing it.

Gender:

The customer base is predominantly Male (57.8%, 104/180), suggesting potential gender-specific product usage or preferences.

Marital Status:

A majority of customers are Partnered (59.4%, 107/180), which indicates fitness and treadmill usage trends among partnered individuals.

Unique Attributes for Categorical Features:

Gender: ['Male', 'Female']

Marital Status: ['Single', 'Partnered']

Product: ['KP281', 'KP481', 'KP781']

This analysis shows that the dataset contains an unbalanced gender distribution (male dominance), a higher proportion of partnered customers, and a varied preference for treadmill models.

Outliers Detection:

Every column has some outliers. ‘Age’, ‘education’ and ‘usage’ have few outliers while ‘income’ and ‘miles’ have more significant number of outliers.

Outliers in Age:

```
Outliers in 'Age':
Outlier Count: 5
     Age  Income  Usage  Fitness  Miles
78    47   56850      4        3     94
79    50   64809      3        3     66
139   48   57987      2        3     64
178   47  104581      4        5    120
179   48   95508      4        5    180
```

Table 2: Outlies in Age

A total of **5 outliers** were identified in the Age variable. These individuals represent extreme age groups, potentially falling outside the main target demographic. These outliers can be reclassified into a separate **"Senior Users"** category to better analyze their unique behaviors and needs.

Outliers in Income:

Outliers in 'Income':
Outlier Count: 19

	Age	Income	Usage	Fitness	Miles
159	27	83416	4	5	160
160	27	88396	4	3	100
161	27	90886	4	4	100
162	28	92131	6	5	180
164	28	88396	6	5	150
166	29	85906	7	5	300
167	30	90886	6	5	280
168	30	103336	5	4	160
169	30	99601	5	5	150
170	31	89641	6	5	260
171	33	95866	4	5	200
172	34	92131	5	5	150
173	35	92131	4	5	360
174	38	104581	5	5	150
175	40	83416	6	5	200
176	42	89641	5	4	200
177	45	90886	5	5	160
178	47	104581	4	5	120
179	48	95508	4	5	180

Table 3: Outliers in Income

19 outliers were detected in the Income variable which indicates a specific group of customers with exceptionally high earnings compared to the rest of the dataset. These individuals could be categorized as **"High-Income Customers"**, as they may represent a premium segment with unique purchasing behaviors and preferences.

```
Outliers in 'Usage':
Outlier Count: 9
     Age  Income  Usage  Fitness  Miles
154   25   70966      6        4    180
155   25   75946      6        5    240
162   28   92131      6        5    180
163   28   77191      7        5    180
164   28   88396      6        5    150
166   29   85906      7        5    300
167   30   90886      6        5    280
170   31   89641      6        5    260
175   40   83416      6        5    200

Outliers in 'Fitness':
Outlier Count: 2
     Age  Income  Usage  Fitness  Miles
14    23   38658      3        1     47
117   31   65220      2        1     21

Outliers in 'Miles':
Outlier Count: 13
     Age  Income  Usage  Fitness  Miles
23    24   44343      5        5    188
84    21   34110      5        4    212
142   22   48556      4        5    200
```

Table 4: Outliers in Usage, Fitness and Miles

Usage Outliers:

A total of **9 outliers** were identified in the Usage variable. The values were significantly higher than the typical range.

Fitness Outliers:

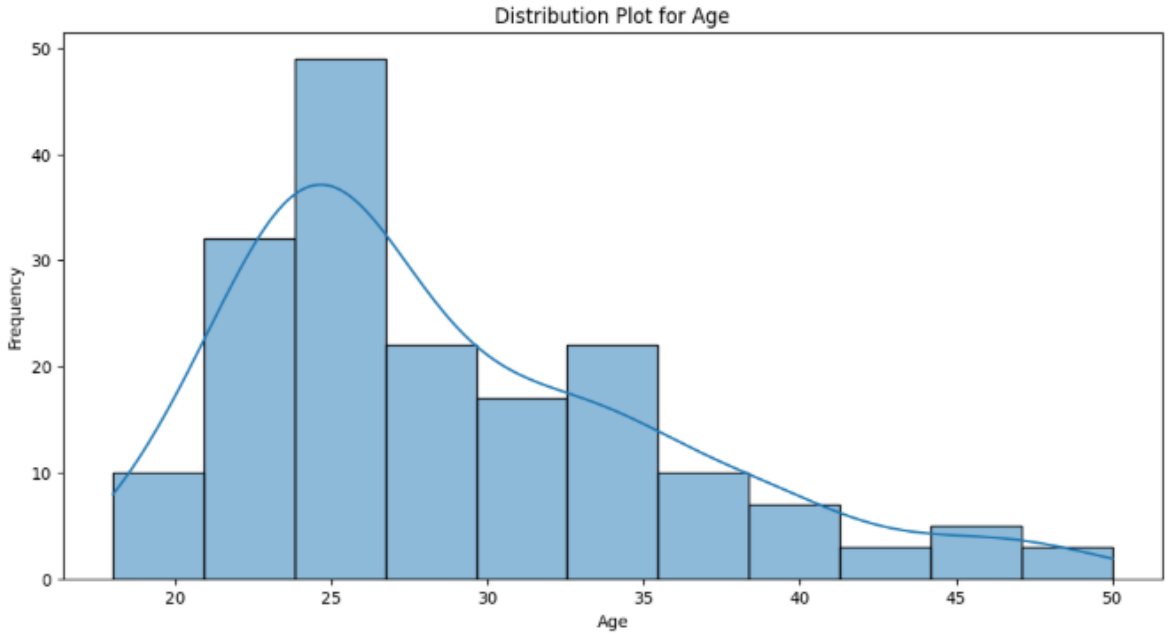
Only **2 outliers** were observed in the Fitness variable, indicating rare cases of individuals with unusually low fitness scores.

Miles Outliers:

13 outliers were detected in the Miles variable, with exceptionally high values. It can be treated as a cluster outlier of high values.

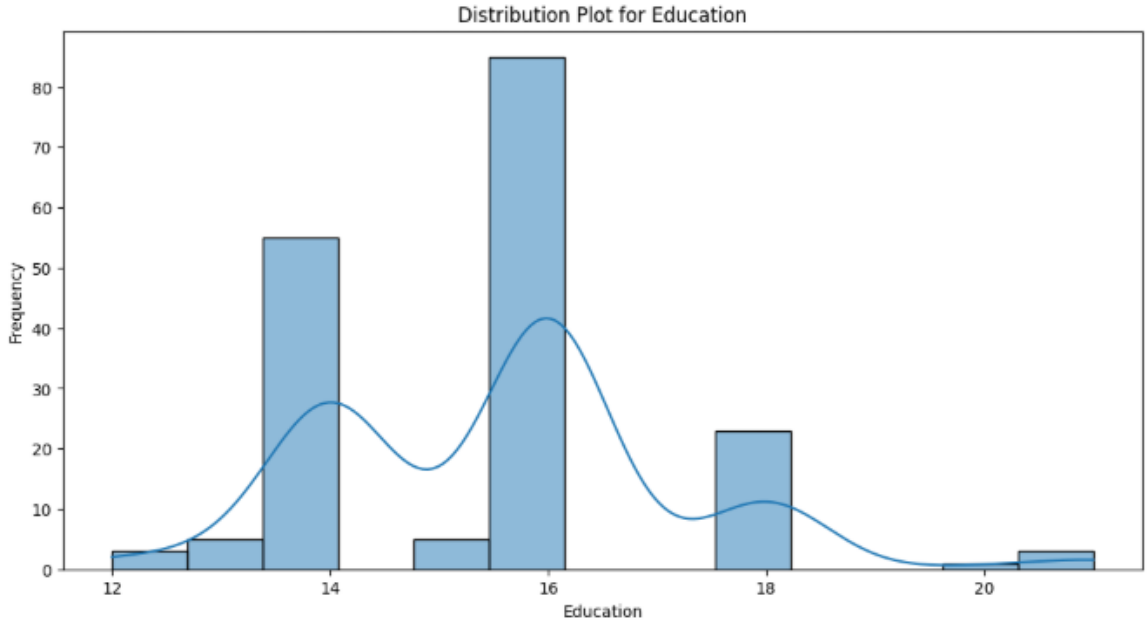
Distribution Plot

Age:



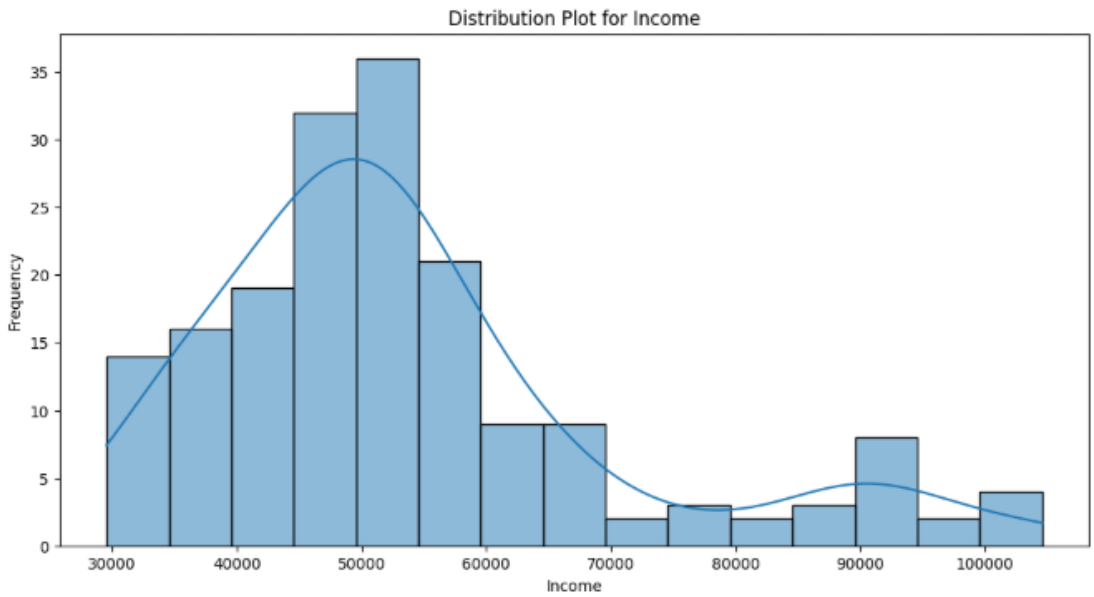
The shape of the distribution shows a right-skewed pattern, where the majority of the individuals fall in the age range of approximately 20-30 years. There are fewer individuals in the higher age groups of above 40.

Education:



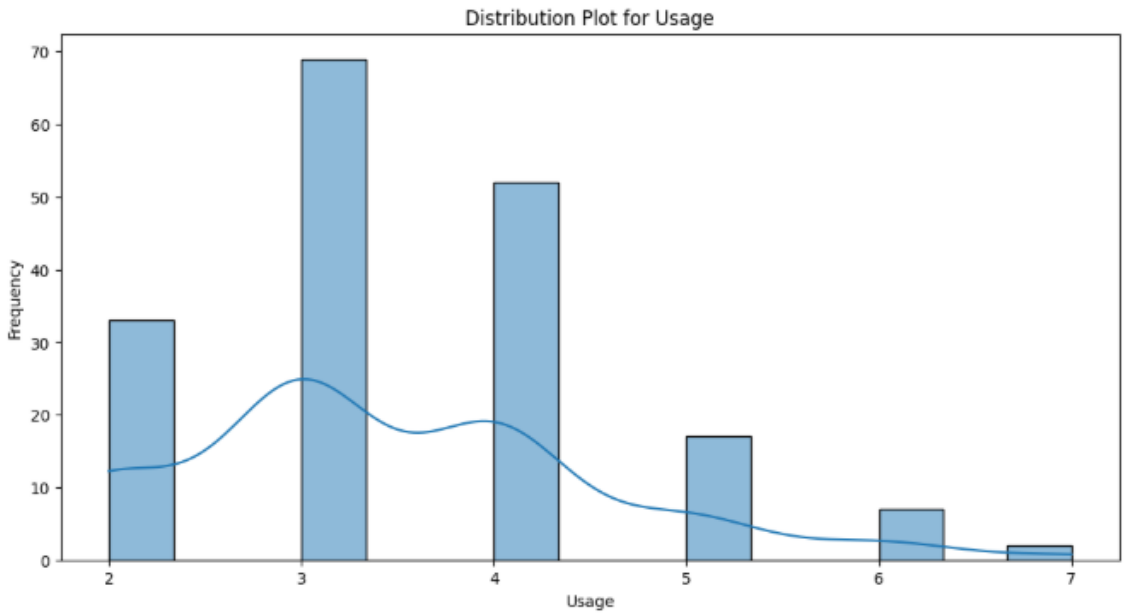
The distribution plot for education shows that the majority of individuals have an education level around 16 years which can link to undergraduate degrees. The frequency declines sharply for education levels beyond 18 years which shows fewer individuals with advanced degrees.

Income:



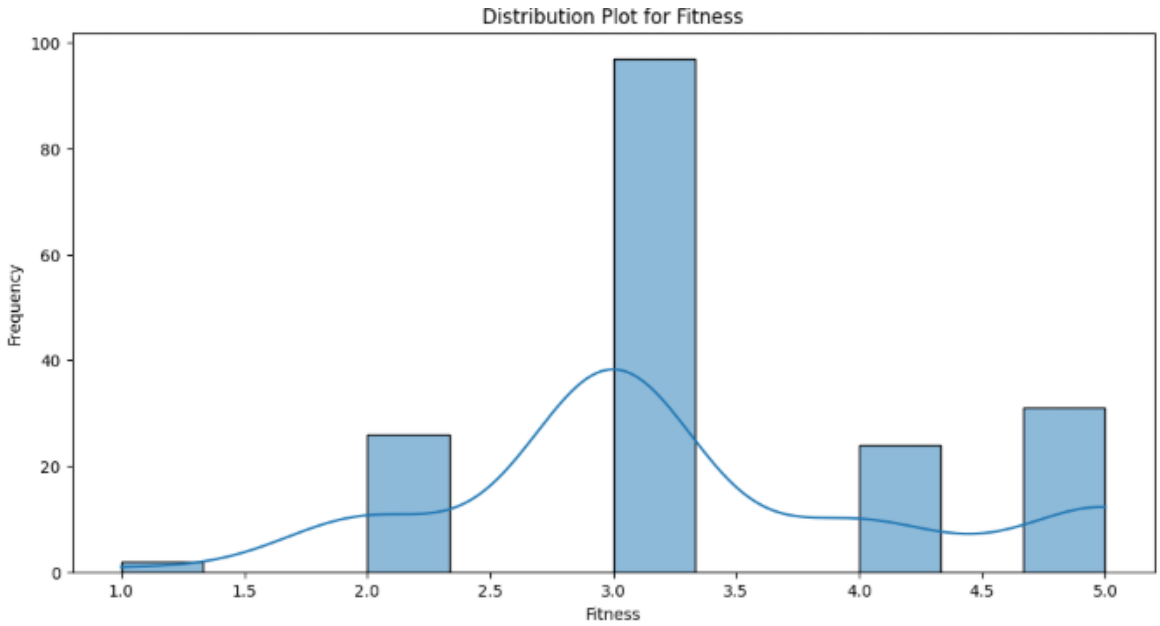
The distribution plot for income shows a concentration of individuals earning between $40,000 and $60,000. The highest frequency is around $50,000. The distribution is slightly right skewed. A smaller proportion of individuals earn higher incomes above $70,000. There is a smaller number of high-income earners exceeding 90,000 $.

Usage:



The distribution plot for usage shows that most individuals use the product or service around 3 to 4 times, with a peak at 3 times. The engagement rate is less at higher usage levels as usage frequencies above 5 are much less common.

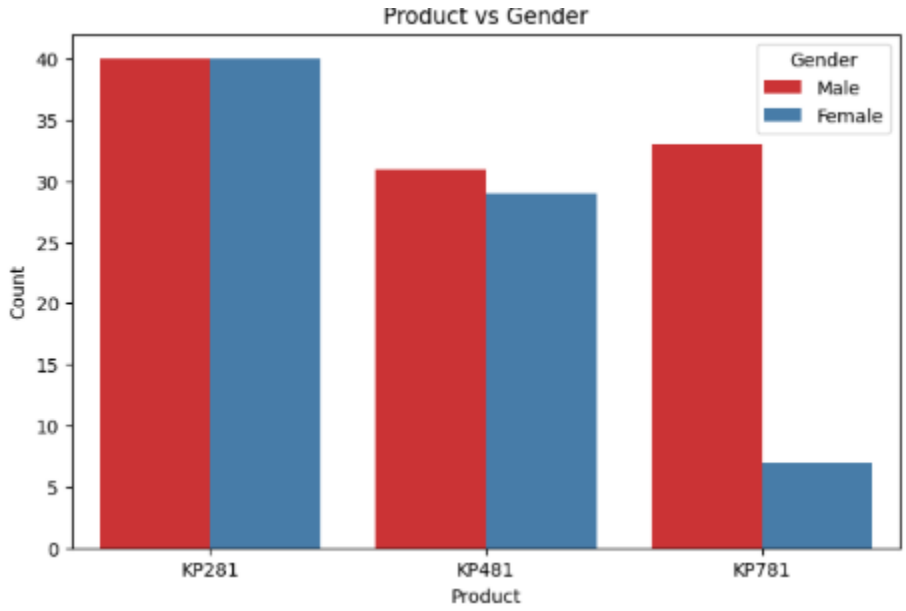
Fitness:



The distribution plot for fitness shows that the majority of individuals have a fitness level centered around 3.5 that indicates an average fitness rating.

Bivariate Analysis

Product vs Gender:

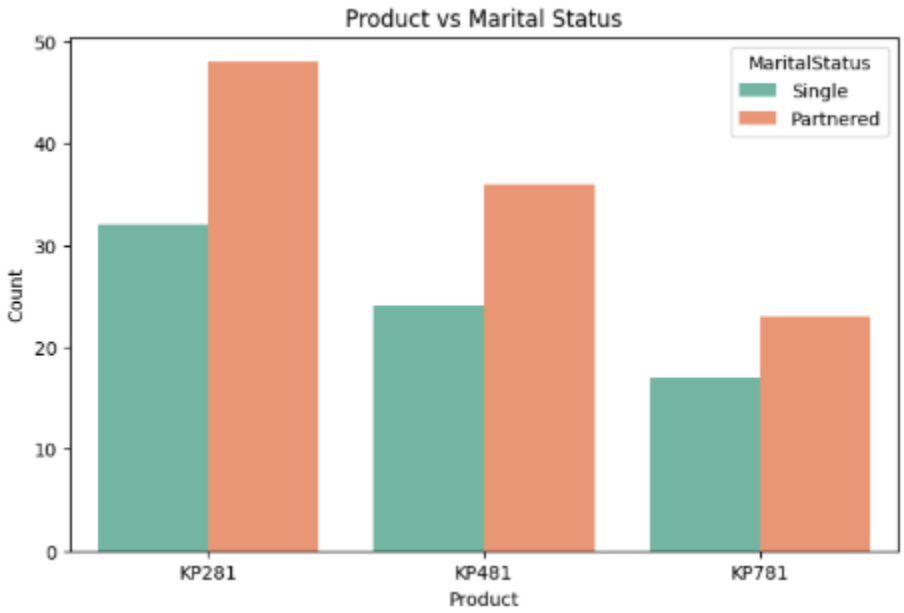


KP281: The distribution of males and females is almost equal.

KP481: Slightly more males than females are associated with this product.

KP781: There is a notable disparity with a strong male preference for this product.

Product Vs Marital Status:



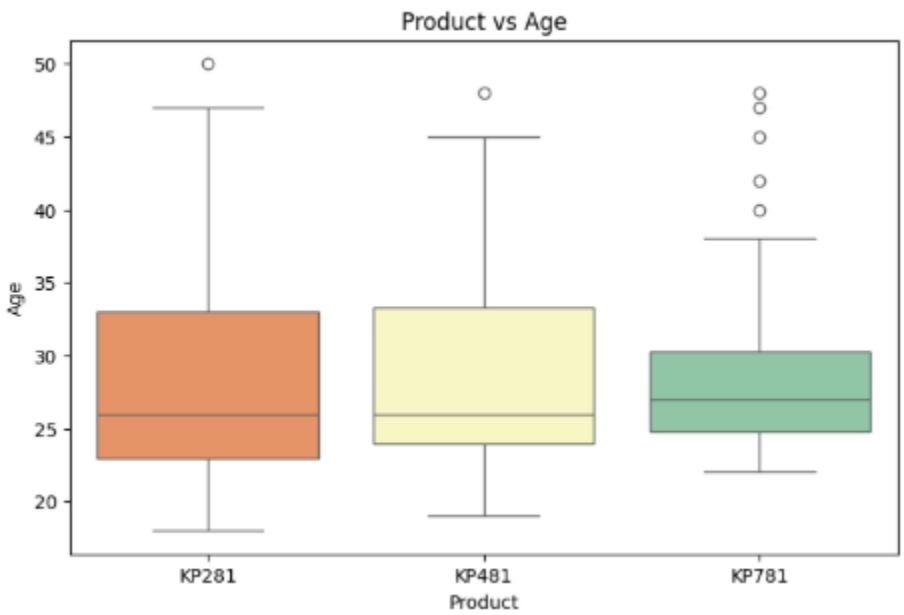
KP281: Partnered individuals have a higher count compared to single individuals.

KP481: Partnered individuals have significantly outnumbered singles.

KP781: The count of partnered individuals is slightly higher than singles, but the difference is less noticeable compared to the other products.

In short, partnered individuals have a higher engagement for all three products, especially KP281 and KP481.

Product Vs Age:



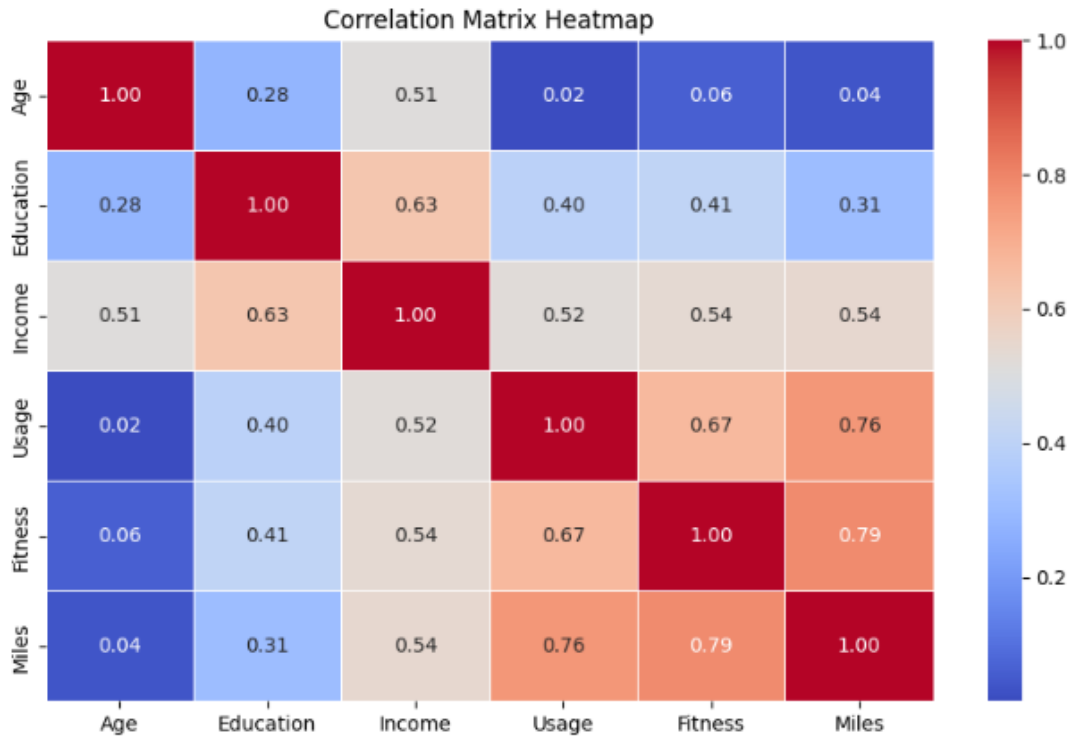
KP281: The median age is around 30 years, and a few outliers above 45 years are observed.

KP481: The median age is similar to KP281, around 30 years. A single outlier is observed above 45 years.

KP781: This product has a slightly younger median age (around 27-28 years). It has a narrower IQR compared to the others. Multiple outliers above 40 years are visible.

KP781 seems to appeal to a younger audience, while KP281 and KP481 have a broader and slightly older age range.

Heatmap:



Strong Correlations:

- Miles and Fitness have a strong positive correlation (0.79). This shows that individuals with better fitness levels tend to cover more miles.
- Miles and Usage (0.76) show a strong positive relationship. This suggests that higher product usage results in greater distances covered.
- Fitness and Usage are positively correlated (0.67), indicating that frequent users of the product tend to have better fitness levels.

Moderate Correlations:

- Income and Education have a moderately positive correlation (0.63). This shows that higher education levels are associated with higher income.
- Income shows moderate correlations with Usage (0.52), Fitness (0.54), and Miles (0.54). This indicates that higher-income individuals may use the product more frequently and achieve better fitness outcomes.

Weak Correlations:

- Age has weak correlations with all other variables, such as Miles (0.04) and Usage (0.02), implying that age does not significantly influence product usage or fitness-related factors.
- Education has a weak correlation with Miles (0.31) and Fitness (0.41)

The heatmap highlights strong interconnections between Usage, Fitness, and Miles, suggesting that engagement with the product positively impacts fitness and distance covered. Age does not drive product engagement or fitness levels significantly.

Conditional Probabilities

Customer Percentage In terms of Procut Purchase

```
Percentage of customers who purchased each product:
```

Product	count
KP281	44.44%
KP481	33.33%
KP781	22.22%

dtype: object

The table shows the percentage of customers who purchased each product:

- KP281 is the most purchased product, accounting for 44.44% of the total purchases.
- KP481 is the second most popular product, with 33.33% of the purchases.
- KP781 is the least purchased product, making up only 22.22% of the total.

KP281 appears to be the most preferred product among customers, while KP781 has the least traction. The marketing strategies need to focus more on KP781 to increase its sales while maintaining the sales of the other two categories.

Product – Gender

57.78% of the treadmill customers are male which indicates that men constitute the majority of the customer base for treadmills. Only 17.50% of KP781 treadmill customers are female suggesting a lower count.

Probability of a Customer Being Female Given that Product is KP281:

The probability of a customer being female, given they purchased the KP281 treadmill, is 50%, indicating an equal likelihood of male and female customers purchasing this product.

The data highlights gender preferences across different treadmill products. Men are the dominant buyers overall but KP281 shows equal gender representation and KP781 skews heavily towards male customers.

Product – Age

The percentage of customers with Age between 20s and 30s among all customers is 60%.

Product – Income

25.00% of treadmill customers are classified as low-income, indicating limited adoption in this segment. Among high-income customers, **64.44%** prefer the KP781 treadmill. This showcases its strong appeal to rich buyers.

The probability of a customer being high-income, given they purchased KP781, is **0.72**. This highlights KP781 as a premium product favored by wealthier individuals.

Product – Fitness

Only 17.22% of the customers have a fitness level of 5 which shows that very few customers have high fitness levels. Customers with a fitness level of 5 show a strong preference for the KP781 treadmill, with 93.55% of them purchasing it. This shows a strong preference for this product among the most fit individuals.

Probability of a Customer with Fitness Level 5 Buying KP781 Treadmill:

The overall probability of a customer both having a fitness level of 5 and buying the KP781 treadmill is 0.94.

The data suggests that the KP781 treadmill is highly popular among customers with the highest fitness level, making it a favored product for wealthy and fit customers. But fitness level 5 represents only 17.22% of customers and marketing campaigns can be directed to capture more elite class customers to increase revenue.

Product - Marital Status

59.44% of treadmill users are partnered. The data shows that partnered individuals make up a significant portion of the customer base for treadmills. This information could be leveraged to target marketing campaigns towards households.

Recommendations

Targeted Marketing Campaigns:

KP281: Market as a versatile product appealing to a broad demographic.

KP781: Focus on high-fitness individuals and males through premium positioning or advanced features.

Gender-Specific Strategies:

Increase KP781's appeal among female customers by addressing their preferences, such as offering tailored features or fitness programs.

Couples and Households:

Develop offers or incentives for partnered customers, such as couple memberships, family-oriented packages, or dual product discounts.

Broaden Accessibility:

Launch new initiatives to appeal KP481 and KP781 to younger or less fit demographics to balance customer distribution and expand the market base.